# RAG, Embeddings, and Vector Databases

Retrieval-Augmented Generation, or RAG, combines information retrieval with a Large Language Model. It allows an application to search an external knowledge source and provide relevant information to the model before an answer is generated.

A RAG knowledge base commonly begins with documents such as text files, PDFs, web pages, or internal documentation. These documents are usually divided into smaller chunks so that relevant sections can be retrieved without sending an entire document to the language model.

An embedding model converts each chunk into a numerical vector that represents aspects of its semantic meaning. A user's question can also be converted into an embedding so that the system can compare the question with stored document chunks.

A vector database stores embeddings and supports similarity search. Technologies such as FAISS, Chroma, Qdrant, Pinecone, Weaviate, Milvus, and pgvector can be used as part of retrieval systems depending on application requirements.

When a user asks a question, the retriever searches for chunks whose vectors are similar to the query vector. The most relevant chunks are then added to the prompt as context for the language model.

Metadata can be stored alongside each document chunk. Information such as the source filename, page number, topic, author, or document category can help identify sources and can also be used to filter retrieval results.

The quality of a RAG system depends heavily on retrieval quality. Chunking strategy, embedding models, search configuration, reranking, document quality, and evaluation all influence whether the language model receives useful context for answering the user's question.